

Kyrillos Samer Mousa

Front-End & Geo-Spatial Solutions Developer | Enterprise ERP & SaaS Systems

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PROFESSIONAL SUMMARY

Front-End & Geo-Spatial Solutions Developer with over 3 years of professional experience in building map-driven applications and large-scale spatial data (Big Data) solutions, currently working on ERP SaaS platforms. Specializes in developing modern front-end interfaces using React.js, Next.js, and TypeScript, while building robust back-end services with Node.js, Express.js, PostgreSQL, and PostGIS. Integrates advanced GIS capabilities through ArcGIS API for JavaScript, Leaflet, and OpenLayers. Works within SaaS and microservices architectures, contributing to system design and applying Design Patterns, Clean Code, and Clean Architecture principles to deliver scalable, maintainable, and high-performance front-end solutions. Passionate about bridging geospatial data with web solutions to build reliable, user-centric digital products that scale effectively and solve complex business challenges.

TECHNICAL SKILLS

Front-End: React.js · Next.js · TypeScript · JavaScript · Redux · TanStack Query · Zustand · Tailwind CSS · Bootstrap

Back-End: Node.js · Express.js · REST API Design · Authentication (JWT / Sessions)

GIS & Spatial: ArcGIS API for JavaScript · Leaflet · OpenLayers · ArcGIS Online · ArcGIS StoryMaps · QGIS

Architecture: Clean Architecture · Design Patterns · Clean Code · SaaS · Microservices · System Design · ERP Systems

Databases: PostgreSQL · PostGIS · Microsoft SQL Server · SQLite

Tools & Workflow: Git · REST APIs · ClickUp · Scrum / Agile · Python · C++ · C# · OOP

PROFESSIONAL EXPERIENCE

Front-End & GIS Developer | BI-Tech

Oct 2025 – Present

- Build scalable enterprise ERP/SaaS front-end using Next.js, React, TypeScript within a microservices architecture.
- Develop back-end services and REST APIs using Node.js and Express.js, integrated with PostgreSQL and PostGIS.
- Implement Clean Architecture, Design Patterns, and Clean Code principles across all modules for long-term maintainability.
- Develop advanced GIS features and interactive map-driven modules using ArcGIS API for JavaScript, Leaflet, and OpenLayers.
- Build dynamic ERP dashboards and document generation (PDF/Excel) using Radix UI and reporting libraries.
- State management with Zustand and TanStack Query for performant, reactive enterprise interfaces.
- Work within an Agile/Scrum environment using ClickUp for project management, sprint planning, and team collaboration.
- Contribute to front-end system design decisions and cross-team collaboration with backend and product.

GIS Front-End Developer | Egyptian Military Survey Authority

2024 – Sep 2025

- Developed large-scale GIS web applications using React.js and OpenLayers for national-level infrastructure projects.
- Built advanced spatial modules: spatial filtering, thematic mapping, buffer zones, coordinate tools, and layer management.
- Processed and visualized Big Data spatial datasets with optimized UI performance for mission-critical environments.
- Integrated front-end with geospatial REST APIs and enterprise backend services.
- Delivered Excel and PDF reporting features for spatial query results and analytical outputs.

Spatial Data Collector | Ministry of Transport – Egypt

Mar 2024

- Collected, validated, and processed spatial data supporting large-scale national transport infrastructure and planning initiatives.
- Ensured data accuracy and consistency for integration into enterprise GIS platforms.

Freelance GIS & Front-End Developer | Self Projects

Ongoing

- Built GIS analytical tools: hydrology analysis, network analysis, topology mapping, and contour digitizing.
- Developed front-end web apps using Node.js, Express.js, and PostgreSQL/PostGIS on the back-end.
- Delivered web apps with dynamic data visualization, interactive dashboards, and reporting (Excel, PDF).

HIGHLIGHTED PROJECTS

Giranha – Enterprise ERP/SaaS Platform

Large-scale ERP SaaS built on microservices. Contributed to system design; implemented Clean Architecture, Design Patterns, and scalable UI modules using Next.js, TypeScript, Zustand, Radix UI, and integrated GIS features. Back-end APIs built with Node.js, Express.js, and PostgreSQL.

Real Estate SaaS Platform

Multi-tenant real estate SaaS with tenant-specific domains and branding, role-based access control (Admin/Manager/Agent), and complete property, client, lead, deal, and requirement workflows. Includes agent assignment to leads/properties and requirement-to-property matching for apartments, villas, land, and studios. AI-powered Arabic natural-language search parses client requirements into structured criteria such as city, district, property type, area, rooms, price, and nearby-location needs, then returns matching properties with match percentages. Map-driven exploration supports manual filters, AI Search, two-way map/list filtering, property markers, category filters, legends, basemap and satellite views, fullscreen, zoom controls, and nearby services such as hospitals, mosques, and churches with distance/buffer-based discovery. Built with Next.js, React, TypeScript, Tailwind CSS, MUI, Zustand, TanStack Query, next-intl, Node.js, Express.js, PostgreSQL/Supabase, and PostGIS.

React GIS Portal

Full-featured web GIS portal with spatial filtering, thematic mapping, buffer zones, routing, and shortest-path network analysis. Optimized for Big Data spatial datasets.

Dashboard & Reporting System

Enterprise dashboard with dynamic charts, KPIs, and automated Excel/PDF report generation for spatial query results and operational data.

KeroStore / Twasul / Shopsy

Front-end e-commerce app, social media platform, and product-listing frontend – all live deployed. Built with React, Next.js, TypeScript, Redux on the front-end and Node.js, Express.js, PostgreSQL on the back-end.

EDUCATION

B.Sc. Surveying, GIS & Remote Sensing | 2019 - 2023

Cairo University – Faculty of Arts | Grade: Very Good

Graduation Project: Geo-Marketing Analytics – Grade: Excellent

COURSES & CERTIFICATIONS

- CS50 – Introduction to Computer Science | Harvard University
- CS50x – Computer Science | HarvardX
- Front-End Development (React, Redux, Next.js, Tailwind) | Route Academy
- Python Programming | ITI – Mahara Tech
- Mastering Python | Self-study
- PostgreSQL & PostGIS | Self-study
- Node.js & Express.js | Self-study
- MS SQL Server | ITI – Mahara Tech
- C++ / OOP / Algorithms / Data Structures / Design Patterns | Self-study
- C# Full Course: OOP, Inheritance, ADO.NET, Advanced Concepts | Self-study

LANGUAGES

Arabic – Native English – Intermediate (B2)